Examiner reports

Q1.

Most students began part (a)(i) successfully forming an equation with V = 0 to achieve the first mark. Having started well, only around half of these students went on to demonstrate the algebraic maturity required to arrive at the given result.

Part (a)(ii) was completed successfully by many students. The majority scored fully marks.

In part (a)(iii) relatively few students identified that T_0 = 38 represented the current year. A significant number of students made no attempt at this question.

The most frequent mark for part (b) was 2/4, with many students starting well and evaluating **both** models at t = 49 or t = 50 and indicating that they were close together. Few students went on to correctly complete a change of sign argument, which would have shown the required result.

An alternative method was to form an equation from the two models and solve numerically using allowed calculator functions. This method was accepted as fully correct provided it was explained thoroughly, as shown in the mark scheme.

Q2.

This question gave students complete choice as to how to go about solving the problem. Many good attempts were seen, with the full range of marks being scored. Different correct approaches were used and given credit. For example, some students modelled using the length and width of the rectangle, others used a radius and angle to form an expression for area using trigonometry.

Marks were often lost through students not fully justifying their answer. Some students failed to identify their variables or were inconsistent in their use. Others did not explain why they had differentiated or how they knew they had found a maximum.

Q6.

The range of marks seen in this question typically varied between up to 4, scored by about 85% of the students, to up to 7 scored by about 25%. Very few completed the whole question successfully, with many making only superficial attempts at part (b).

Part (a)(i) and (ii). Most students answered correctly with only the occasional arithmetic error.

Part (a)(iii). Many students scored 2 out of the 3 marks available; although they understood what to do and followed correct manipulation, taking logarithms correctly, they didn't respond to the request for integers in the final expression. Those students who didn't simplify the expressions in their working often made an error.

Part (b)(i). Students were expected to attempt the required differentiation using the quotient or chain rule. Those that did this often made a sign error or had a wrong coefficient, sometimes including a t in their expression. Although some excellent algebra was seen in deriving the given result using the initial expression for N, many were unsuccessful in eliminating the $9e^{\frac{t}{8}}$ term, and some gave up without trying. However, many students tried to rearrange the given equation for N to make $e^{-\frac{t}{8}}$ or t the subject, and then differentiate. It is difficult to see why they considered this approach would be helpful, but many made an algebraic error and little progress. Some tried to solve the differential equation and made little or no progress.

Part (b)(ii). Most students who attempted this understood the question to mean finding the maximum value of N rather than that of the growth rate $\frac{dN}{dt}$, but most had to abandon their attempt. Of those who did differentiate the growth rate expression, most wrote $\frac{d^2N}{dt^2}$, which was condoned; if their attempt led to the maximum growth rate occurring when N = 250 they usually completed successfully; those who had worked with inequalities were condoned. Some students thought T = 250 was the required answer. Few students saw that the maximum value of the growth rate was at N = 250 by inspection, (due to the symmetries of the quadratic function involved) but some concise and corrects solutions to the problem were seen.

Q7.

In part (a)(i), some students merely doubled the given equation and then divided by 2; this did not convince examiners and earned no marks. Students were expected to show the separate areas of the faces of the cuboid; the minimum working required was sight of $2xy + 6xy$ as well as $6x^2$, before combining these to give $6x^2 + 8xy = 32$ and hence the printed result. Some proved the result by considering half the surface area and, provided enough explanation was given, this also scored full marks.

In part (a)(ii), the majority earned a single mark for writing down a correct expression for the volume in terms of x and y. The algebra required to rearrange the equation from part (a)(i) to make y or xy the subject proved too difficult for many and once again highlighted students' algebraic weaknesses as they tried to reach the printed answer but violated various algebraic laws on the way.

In part (b), whilst the term $12x$ was almost invariably differentiated correctly, the same could not be said of the second term in the expression for V, and $-27x^2$ was a common incorrect answer.

In part (c)(i), although most students realised the need to substitute $x = \frac{4}{3}$ into their expression for $\frac{dy}{dx}$, not all showed the required working to prove that $\frac{dy}{dx} = 0$. It was not sufficient to write $12 - \frac{27\left(\frac{4}{3}\right)^2}{4} = 12 - 12 = 0$; evidence of correct manipulation of the fractions was needed. Once again, many students did not write a concluding statement with regard to the zero gradient implying that there was a stationary point.

In part (c)(ii), by far the most common error when finding $\frac{d^2y}{dx^2}$ was the omission of the minus sign. The final mark was still available even if students made errors in finding the second derivative, provided the substitution of $x = \frac{4}{3}$ was done correctly and the appropriate conclusion was drawn regarding the nature of the stationary point.

Q8.

In part (a) finding $\frac{dV}{dt}$ caused few problems apart from those who chose to multiply by 4 before

differentiating. Some will insist on adding $+c$ to their derivative despite advice given in previous reports.

In (b)(i) the concept of "rate of change" seemed to be better understood this time, helped by there being only one derivative to choose from; careless arithmetic when trying to simplify
0.75 – 3 spoiled many solutions.
Part (b)(ii) Most candidates realised that the volume was decreasing when t = 1 but many failed to give an adequate reason; most neglected to state the crucial fact that the rate of change was negative.

In part (c)(i) candidates were expected to solve an equation of the form $\frac{3}{4}t^2 - 3 = 0$ but most
chose to guess that t = 2 was the solution and attempted to verify this fact.
Part (c)(ii) Once again the explanation as to why the stationary value was a minimum value was inadequate, very often taking no account of the value of t found in the previous part of the

question. Some candidates who correctly found that the second derivative was $\frac{3t}{2}$ and

proceeded to solve the equation $\frac{3t}{2} = 0$ often obtained a value such as 6 and stated that this
positive solution indicated a minimum value.

Q9.

In part (a), almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

In part (b), those who substituted t = 2 into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt}$ = 0 is the condition for a stationary point. Some **assumed** that a stationary point occurred when t = 2, went straight to the test for maximum or minimum and only scored half the marks.

It was advisable to use the second derivative test here; those who considered values of $\frac{dy}{dt}$ on either side of t = 2 usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c)(i), the concept of 'rate of change' was not understood by many who failed to realise the need to substitute t = 1 into $\frac{dy}{dt}$. Some candidates wrongly substituted t = 1 into the initial expression for y or into their expression for $\frac{d^2y}{dt^2}$ and these candidates were unable to score any marks at all on this part. Even those who used $\frac{dy}{dt}$ sometimes made careless arithmetic errors when adding three numbers.

In part (c)(ii), it was not enough to simply write the word "increasing": some explanation about $\frac{dy}{dt}$ being positive was also required. Some candidates erroneously found the value of the second derivative when t = 1 or calculated the value of y on either side of t = 1.

Q10.

Part (a)(i) Usually after a few abortive attempts many candidates realised that they had to add together the areas of the various faces. Once they had the correct expression for the total surface area most candidates were able to obtain the printed result. There was clearly some fudging on the part of weaker candidates and they could earn little more than a single method mark.

Part (a)(ii) This was surprisingly one of the biggest discriminators on the paper with only the best candidates being able to obtain the correct expression for V. Trying to make y the subject of the equation from part (a)(i) caused problems for many who took this approach; others substituted for xy in the formula $V = 6x(xy)$ and were often more successful. Once again it was quite common to see totally incorrect expressions being miraculously transformed into the printed answer.

Part (b)(i) Most candidates scored full marks for this basic differentiation although it was not always clearly identified as $\frac{dV}{dx}$ in their working.

Part (b)(ii) Most substituted x = 2 into their expression for $\frac{dV}{dx}$ and found the value to be zero. It was then necessary to make a statement about the implication of there being a stationary value in order to score full marks.

Part (c) Some made a sign error when finding the second derivative, but the majority of candidates scored full marks in this part. Credit was given if the correct conclusion was drawn from the sign of their second derivative, provided no further arithmetic errors occurred.

Q11.

This opening question provided a confident start for most candidates, with correct answers often seen to parts (a) and (b)(i). However, part (b)(ii) was answered relatively